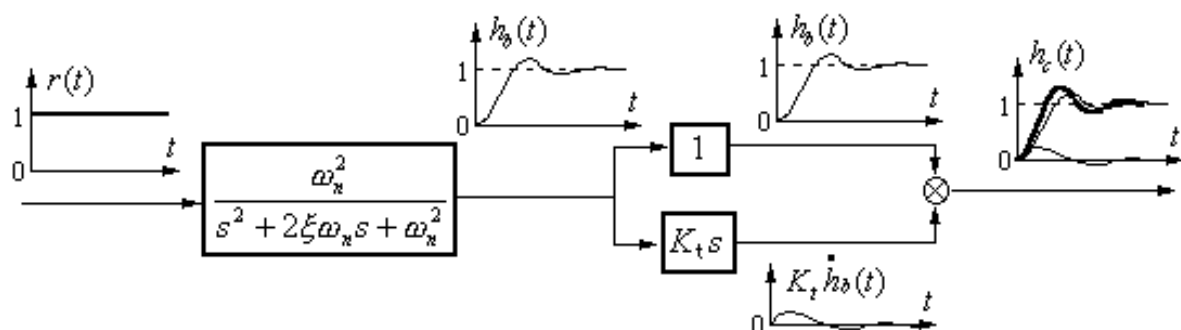
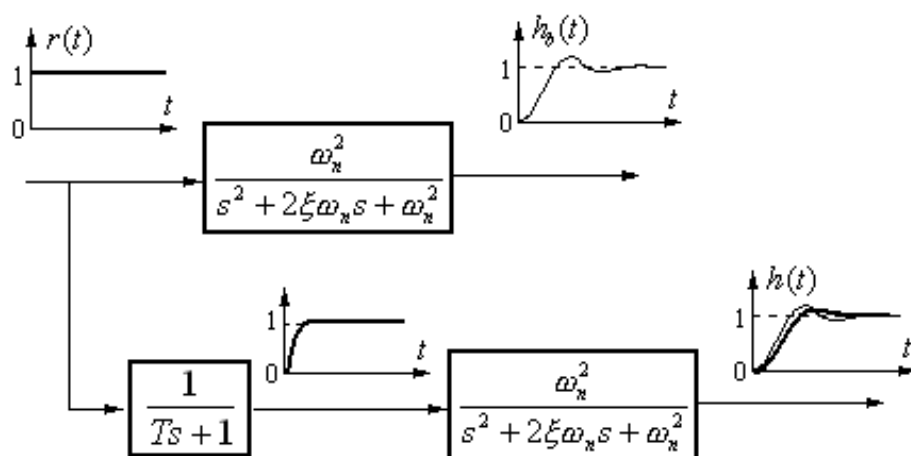


- 附加**闭环零/极点**对系统性能的影响



附加闭环零点时系统响应合成示意图



附加闭环极点时系统响应合成示意图

§ 3.4 高阶系统的阶跃响应及动态性能

§ 3.4.1 高阶系统单位阶跃响应

$$\Phi(s) = \frac{M(s)}{D(s)} = \frac{b_m s^m + b_{m-1} s^{m-1} + \cdots + b_1 s + b_0}{a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0}$$

$$= \frac{K \prod_{i=1}^m (s - z_i)}{\prod_{j=1}^n (s - \lambda_j)} \quad n \geq m$$

$$C(s) = \Phi(s) \cdot \frac{1}{s} = \frac{K \prod_{i=1}^m (s - z_i)}{s \prod_{j=1}^n (s - \lambda_j)}$$

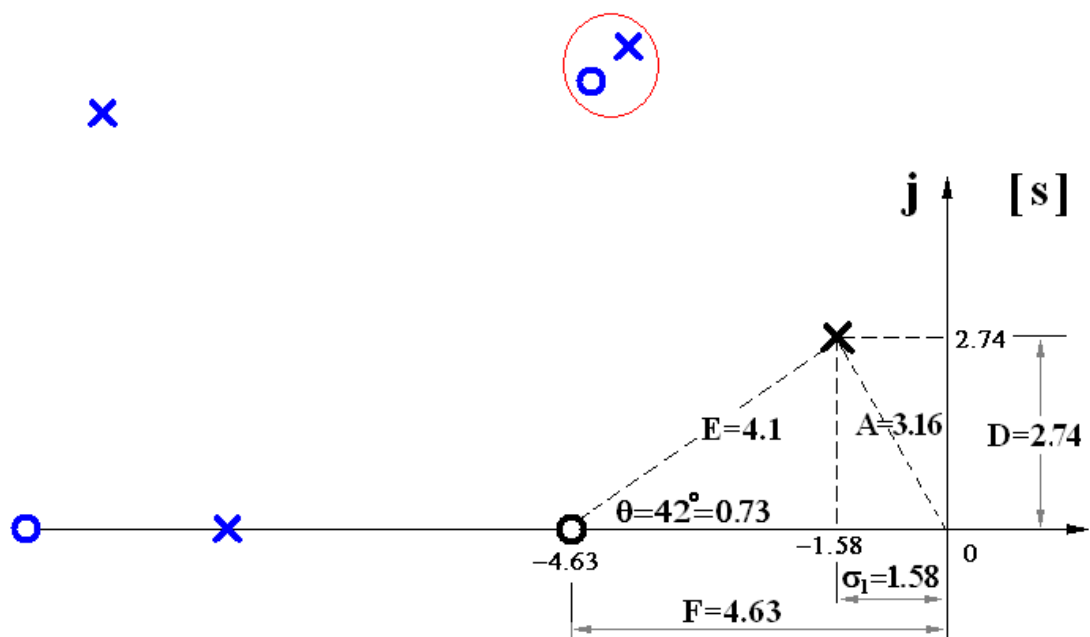
$$= \frac{C_0}{s} + \frac{C_1}{s - \lambda_1} + \cdots + \frac{C_n}{s - \lambda_n} = \frac{C_0}{s} + \sum_{j=1}^n \frac{C_j}{s - \lambda_j}$$

$$= \frac{M(0)}{D(0)} \cdot \frac{1}{s} + \sum_{j=1}^n \frac{M(s)}{s D'(s)} \bigg|_{s=\lambda_j} \frac{1}{s - \lambda_j}$$

$$c(t) = C_0 + \sum_{j=1}^n C_j e^{\lambda_j t}$$

$$= \frac{M(0)}{D(0)} + \sum_{j=1}^n \frac{M(s)}{s D'(s)} \bigg|_{s=\lambda_j} \cdot e^{\lambda_j t}$$

$$= \Phi(0) + \sum_{\lambda_i = -\alpha_i} \frac{M(s)}{s D'(s)} \bigg|_{s=-\alpha_i} \cdot e^{-\alpha_i t} + \sum_{\lambda_i = -\sigma_i \pm j\omega_{di}} A_i e^{-\sigma_i t} \sin(\omega_{di} t + \varphi_i)$$



§ 3.4.2 闭环主导极点

主导极点：距离虚轴最近而且附近又没有闭环零点的闭环极点

§ 3.4.3 估算高阶系统动态性能指标的零点极点法

- (1) $\Phi(s) \Rightarrow$ 闭环零极点图；
- (2) 略去非主导零极点和不非常靠近虚轴的“偶极子”，保留主导极点；
- (3) 按 P73 表 3-7 相应公式估算系统动态性能。

$$G(s) = \frac{K(\tau_1 s + 1) \dots (\tau_m s + 1)}{s^v (T_1 s + 1) \dots (T_{n-v} s + 1)} = \frac{K(b_m s^m + \dots + b_1 s + 1)}{a_n s^n + \dots + a_{v+1} s^{v+1} + s^v}$$

$$\Phi(s) = \frac{K(b_m s^m + \dots + b_1 s + 1)}{a_n s^n + \dots + a_{v+1} s^{v+1} + s^v + K(b_m s^m + \dots + b_1 s + 1)}$$

$$\begin{aligned} D(s) &= a_n s^n + \dots + a_{v+1} s^{v+1} + s^v + K(b_m s^m + \dots + b_1 s + 1) = 0 \\ &= a_n s^n + \dots + (a_k + K b_k) s^k + \dots + K = 0 \end{aligned}$$